

SIMATS ENGINEERING

CO1 - ASSESSMENT TOOL 2

ERROR ANALYSIS TASK

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COURSE NAME: Deep Learning
FACULTY : Dr.Arul Raj A.M

COURSE CODE: MLA0403

COURSE OUTCOME COVERED

CO 1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BTL-4)

CO1 Weightage: 15%

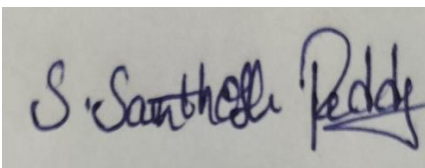
Assessment Tool Weightage: 30%

Duration: 30 minutes

Marks: 25

Code of Conduct:

I **S. Santhosh Reddy (192425017)**(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

QUESTIONS:

Total Marks:25

1. Error Analysis in Maximum Likelihood Estimation (MLE)

A student builds a binary classifier using Maximum Likelihood Estimation. The likelihood function is incorrectly written as the sum of probabilities instead of the product. The model produces poor parameter estimates even with large data.

Tasks:

- a) Identify the conceptual error in the likelihood formulation.

Answer :

Conceptual error

The likelihood was incorrectly written as a sum of probabilities instead of the product of probabilities. Likelihood must represent the joint probability of all independent samples, which is multiplicative, not additive.

- b) Explain how this mistake affects parameter estimation.

Answer :

Using a sum breaks the probabilistic foundation of MLE. It no longer reflects the true probability distribution, leading to biased and inconsistent parameter estimates, even with large datasets.

- c) Suggest the correct mathematical formulation and reasoning.

Answer :

For independent samples $\{x_1, x_2, \dots, x_n\}$, the likelihood is:

$$L(\theta) = \prod_{i=1}^n P(x_i | \theta)$$

This ensures parameters maximize the probability of observing the entire dataset.

d) How would using log-likelihood fix computational issues?

Answer :

Taking logs converts the product into a sum:

$$\ell(\theta) = \sum_{i=1}^n \log P(x_i | \theta)$$

This avoids numerical underflow, simplifies differentiation, and makes optimization computationally efficient.

2. Error Diagnosis in Perceptron Learning

While implementing a Perceptron, a student observes that the model fails to converge even after many iterations on a linearly separable dataset.

Tasks:

a) List possible implementation errors in the learning algorithm.

Answer :

- ☐ Missing bias term in updates
- ☐ Incorrect weight update rule (wrong sign or formula)
- ☐ Not shuffling data between epochs
- ☐ Improper stopping condition

b) Analyze how incorrect learning rate or weight updates affect convergence.

Answer :

- ☐ Too high $\rightarrow$ oscillations, divergence
- ☐ Too low $\rightarrow$ extremely slow convergence
- ☐ Incorrect update $\rightarrow$ weights fail to move toward separating hyperplane

c) Identify dataset-related issues that may cause this problem.

Answer :

- ☐ Data not linearly separable
- ☐ Mislabelled samples
- ☐ Duplicate or redundant points causing instability

d) Propose modifications to ensure convergence.

Answer :

- ☐ Use appropriate learning rate scheduling
- ☐ Add bias term explicitly
- ☐ Normalize input features
- ☐ Switch to more robust algorithms (e.g., margin-based Perceptron, SVM)

3. Backpropagation Gradient Error Analysis

A Multilayer Perceptron trained using Backpropagation shows increasing loss during training instead of decreasing.

Tasks:

a) Identify possible mathematical or coding errors in gradient computation.

Answer :

- ☐ Wrong derivative of activation function
- ☐ Forgetting chain rule in backpropagation
- ☐ Sign errors in weight updates
- ☐ Incorrect indexing in matrix operations

b) Explain the role of activation functions in gradient flow.

Answer :

Activation functions regulate gradient flow. Poor choices (e.g., sigmoid in deep layers) can cause vanishing gradients, while ReLU maintains stronger gradient signals.

c) Analyze how vanishing or exploding gradients affect training.

Answer :

- ☐ Vanishing → weights stop updating, training stalls
- ☐ Exploding → unstable updates, loss diverges

d) Suggest techniques to fix the issue (e.g., normalization, initialization).

Answer :

- ☐ Use ReLU or Leaky ReLU
- ☐ Apply Batch Normalization
- ☐ Use proper initialization (Xavier/He)
- ☐ Gradient clipping to control exploding gradients

4. Gradient Descent vs SGD Error Investigation

A model trained using Gradient Descent converges very slowly, while the same model with Stochastic Gradient Descent shows unstable loss fluctuations.

Tasks:

a) Explain why batch gradient descent is slow in large datasets.

Answer :

It computes gradients over the entire dataset per step, which is computationally expensive for large datasets.

b) Analyze the cause of fluctuations in SGD.

Answer :

SGD updates weights using one sample at a time, introducing high variance in gradient estimates, leading to noisy loss curves.

c) Compare the error behavior of Mini-Batch Gradient Descent.

Answer :

- ☐ Less noisy than SGD
- ☐ Faster than full batch GD
- ☐ Provides smoother convergence

d) Recommend the best approach and justify with error trade-offs.

Answer :

Mini-Batch Gradient Descent with learning rate scheduling. It offers the best trade-off between speed, stability, and generalization.

5. Curse of Dimensionality Error Analysis

A machine learning model built using high-dimensional data performs well on training data but poorly on test data due to overfitting, related to the Curse of Dimensionality.

Tasks:

a) Explain how high dimensionality increases model error.

Answer :

As dimensions grow, data becomes sparse. Models overfit training data because distances lose meaning, increasing generalization error.

b) Identify signs of overfitting in this scenario.

Answer :

- ☐ High training accuracy
- ☐ Low test accuracy
- ☐ Large gap between training and validation performance

c) Analyze why distance-based learning fails in high dimensions.

Answer :

In high dimensions, distances between points become nearly uniform, making nearest-neighbor or clustering methods ineffective.

d) Suggest dimensionality reduction or regularization techniques to improve performance.

Answer :

- ☐ Dimensionality reduction (PCA, t-SNE, Autoencoders)
- ☐ Regularization (Dropout, L2 penalty)
- ☐ Feature selection to retain only informative variables

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach

Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand
		solution			